# Robot Simulator - Project Report

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## 1. GitHub Repository URL

**Repository:** `https://github.com/kabdulrahim/robot-project`

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## 2. Project Overview

The Robot Simulator is a Java application that simulates a robot walking around a room. The robot holds a pen that can be raised or lowered. When the pen is down, the robot traces its path on an N×N floor grid as it moves.

### Technology Stack

- **Language:** Java 11+
- **Build Tool:** Maven 3.6+
- **Testing Framework:** JUnit 5
- **Version Control:** Git/GitHub

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## 3. User Stories and Requirements

### User Stories

| ID | User Story |
|---|---|
| US1 | As a user, I want to initialize a floor of any size so that I can simulate the robot on different room sizes. |
| US2 | As a user, I want to control the pen state so that I can choose when the robot draws. |
| US3 | As a user, I want to turn the robot left or right so that I can change its direction. |
| US4 | As a user, I want to move the robot forward so that it can traverse the floor. |
| US5 | As a user, I want to see the current robot status so that I know its position, direction, and pen state. |
| US6 | As a user, I want to print the floor so that I can see the traced pattern. |
| US7 | As a user, I want to replay command history so that I can review my actions. |
| US8 | As a user, I want to quit the program gracefully. |

### Functional Requirements

| Req ID | Requirement Description | Related User Story |
|---|---|---|
| **R1** | The system shall initialize an N×N floor grid where N is a positive integer provided by the user. | US1 |
| **R2** | Upon initialization, the robot shall be positioned at coordinates (0, 0), facing north, with pen up. | US1 |
| **R3** | The system shall allow the user to raise the pen (pen up) so the robot moves without drawing. | US2 |
| **R4** | The system shall allow the user to lower the pen (pen down) so the robot draws while moving. | US2 |
| **R5** | The system shall allow the user to turn the robot 90° clockwise (right). | US3 |
| **R6** | The system shall allow the user to turn the robot 90° counter-clockwise (left). | US3 |
| **R7** | The system shall move the robot forward by a specified number of spaces in the current direction. | US4 |
| **R8** | When the pen is down, the system shall mark all floor cells along the robot's path (set to 1). | US4 |
| **R9** | The system shall prevent the robot from moving beyond floor boundaries. | US4 |
| **R10** | The system shall display the robot's current position, pen state, and facing direction. | US5 |
| **R11** | The system shall print the floor grid showing asterisks (*) for marked cells and blanks for unmarked cells. | US6 |
| **R12** | The system shall maintain a history of executed commands (excluding P, H, Q). | US7 |
| **R13** | The system shall replay all commands from history when requested. | US7 |
| **R14** | The system shall terminate gracefully when the quit command is issued. | US8 |
| **R15** | All commands shall be case-insensitive (e.g., 'U' and 'u' are equivalent). | All |
| **R16** | The system shall provide meaningful error messages for invalid commands or operations. | All |

## 4. Command Reference

| Command | Description | Example |
|---|---|---|
| `I n` | Initialize n×n floor | `I 10` |
| `U` | Pen up (stop drawing) | `U` |
| `D` | Pen down (start drawing) | `D` |

| Command | Description | Example |
|---|---|---|
| `R` | Turn right (clockwise) | `R` |
| `L` | Turn left (counter-clockwise) | `L` |
| `M s` | Move forward s spaces | `M 5` |
| `P` | Print floor grid | `P` |
| `C` | Show current status | `C` |
| `H` | Replay command history | `H` |
| `Q` | Quit program | `Q` |

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## 5. Screenshots of Each Function

### 5.1 Program Startup

```
Robot Simulator
Commands: I n (initialize), U (pen up), D (pen down), R (turn right),
          L (turn left), M s (move), P (print), C (status), H (history), Q (quit)

Enter command:
```

**Requirement Fulfilled:** Program launches successfully

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### 5.2 Initialize Floor (R1, R2)

```
Enter command: I 10
Initialized 10 x 10 floor

Enter command: C
Position: 0, 0 - Pen: up - Facing: north
```

**Requirements Fulfilled:** R1 (10×10 floor created), R2 (robot at origin, facing north, pen up)

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### 5.3 Pen Down (R4)

```
Enter command: D
Pen is now down

Enter command: C
Position: 0, 0 - Pen: down - Facing: north
```

**Requirement Fulfilled:** R4 (pen lowered)

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### 5.4 Move Forward with Pen Down (R7, R8)

```
Enter command: M 4
Moved 4 spaces

Enter command: C
Position: 0, 4 - Pen: down - Facing: north
```

**Requirements Fulfilled:** R7 (moved 4 spaces), R8 (path marked)

---

### 5.5 Turn Right (R5)

```
Enter command: R
Turned right

Enter command: C
Position: 0, 4 - Pen: down - Facing: east
```

**Requirement Fulfilled:** R5 (turned 90° clockwise)

---

### 5.6 Move and Draw Horizontal Line (R7, R8)

```
Enter command: M 3
Moved 3 spaces

Enter command: C
Position: 3, 4 - Pen: down - Facing: east
```

**Requirements Fulfilled:** R7, R8 (horizontal line drawn)

---

### 5.7 Print Floor (R11)

```
Enter command: P
  9
  8
  7
  6
  5
  4 * * * *
  3 *
```

```
  2 *
  1 *
  0 *
    0 1 2 3 4 5 6 7 8 9
```

**Requirement Fulfilled:** R11 (asterisks for marked cells, blanks for unmarked)

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### 5.8 Pen Up (R3)

```
Enter command: U
Pen is now up

Enter command: M 2
Moved 2 spaces

Enter command: P
  9
  8
  7
  6
  5
  4 * * * *
  3 *
  2 *
  1 *
  0 *
    0 1 2 3 4 5 6 7 8 9
```

**Requirement Fulfilled:** R3 (moved without drawing - no new marks)

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### 5.9 Turn Left (R6)

```
Enter command: L
Turned left

Enter command: C
Position: 5, 4 - Pen: up - Facing: north
```

**Requirement Fulfilled:** R6 (turned 90° counter-clockwise)

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### 5.10 Boundary Protection (R9)

```
Enter command: I 5
Initialized 5 x 5 floor
```

```
Enter command: M 10
Error: Cannot move beyond floor boundary. Stopped at position (0, 4)
```

**Requirement Fulfilled:** R9 (prevented out-of-bounds movement)

---

### 5.11 Command History Replay (R12, R13)

```
Enter command: I 5
Initialized 5 x 5 floor

Enter command: D
Pen is now down

Enter command: M 3
Moved 3 spaces

Enter command: H
=== Replaying History ===
> I 5
Initialized 5 x 5 floor
> D
Pen is now down
> M 3
Moved 3 spaces
=== End of Replay ===
```

**Requirements Fulfilled:** R12 (history maintained), R13 (replay works)

---

### 5.12 Case Insensitivity (R15)

```
Enter command: i 5
Initialized 5 x 5 floor

Enter command: d
Pen is now down

Enter command: m 2
Moved 2 spaces

Enter command: c
Position: 0, 2 - Pen: down - Facing: north
```

**Requirement Fulfilled:** R15 (lowercase commands work)

---

### 5.13 Error Handling (R16)

```
Enter command: X
Error: Unknown command 'X'. Valid commands: U, D, R, L, M, P, C, Q, I, H

Enter command: M
Error: Move command requires a number of spaces (e.g., 'M 5')

Enter command: I 0
Error: Floor size must be at least 1
```

**Requirement Fulfilled:** R16 (meaningful error messages)

---

### 5.14 Quit Program (R14)

```
Enter command: Q
Goodbye!
```

**Requirement Fulfilled:** R14 (graceful termination)

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## 6. Requirements to Unit Test Mapping

### 6.1 Direction Tests (DirectionTest.java)

| Test Case | Description | Requirement(s) | Result |
|---|---|---|---|
| testNorthDisplayName | NORTH displays as "north" | R10 | ☑ PASS |
| testEastDisplayName | EAST displays as "east" | R10 | ☑ PASS |
| testSouthDisplayName | SOUTH displays as "south" | R10 | ☑ PASS |
| testWestDisplayName | WEST displays as "west" | R10 | ☑ PASS |
| testTurnRightFromNorth | NORTH → EAST when turning right | R5 | ☑ PASS |
| testTurnRightFromEast | EAST → SOUTH when turning right | R5 | ☑ PASS |
| testTurnRightFromSouth | SOUTH → WEST when turning right | R5 | ☑ PASS |
| testTurnRightFromWest | WEST → NORTH when turning right | R5 | ☑ PASS |
| testTurnLeftFromNorth | NORTH → WEST when turning left | R6 | ☑ PASS |
| testTurnLeftFromWest | WEST → SOUTH when turning left | R6 | ☑ PASS |
| testTurnLeftFromSouth | SOUTH → EAST when turning left | R6 | ☑ PASS |
| testTurnLeftFromEast | EAST → NORTH when turning left | R6 | ☑ PASS |

| Test Case | Description | Requirement(s) | Result |
|---|---|---|---|
| testFourRightTurnsReturnToOriginal | 4 right turns = full circle | R5 | ☑ PASS |
| testFourLeftTurnsReturnToOriginal | 4 left turns = full circle | R6 | ☑ PASS |
| testNorthDelta | NORTH has delta (0, 1) | R7 | ☑ PASS |
| testEastDelta | EAST has delta (1, 0) | R7 | ☑ PASS |
| testSouthDelta | SOUTH has delta (0, -1) | R7 | ☑ PASS |
| testWestDelta | WEST has delta (-1, 0) | R7 | ☑ PASS |
| testAllDirectionsHaveDisplayNames | All directions have names | R10 | ☑ PASS |
| testTurnRightThenLeft | Right then left = original | R5, R6 | ☑ PASS |
| testTurnLeftThenRight | Left then right = original | R5, R6 | ☑ PASS |

**Total: 30 tests (including parameterized variations)**

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### 6.2 Robot Tests (RobotTest.java)

| Test Case | Description | Requirement(s) | Result |
|---|---|---|---|
| testInitialPosition | Robot starts at (0, 0) | R2 | ☑ PASS |
| testInitialDirection | Robot starts facing north | R2 | ☑ PASS |
| testInitialPenState | Robot starts with pen up | R2 | ☑ PASS |
| testPenDown | penDown() sets pen to down | R4 | ☑ PASS |
| testPenUp | penUp() sets pen to up | R3 | ☑ PASS |
| testTurnRight | turnRight() changes direction clockwise | R5 | ☑ PASS |
| testTurnLeft | turnLeft() changes direction counter-clockwise | R6 | ☑ PASS |
| testSetX | setX() updates X position | R7 | ☑ PASS |
| testSetY | setY() updates Y position | R7 | ☑ PASS |
| testSetDirection | setDirection() updates direction | R5, R6 | ☑ PASS |

| Test Case | Description | Requirement(s) | Result |
|---|---|---|---|
| testReset | reset() restores initial state | R2 | ☑ PASS |
| testGetStatusPenUp | Status shows "Pen: up" | R10 | ☑ PASS |
| testGetStatusPenDown | Status shows "Pen: down" | R10 | ☑ PASS |
| testGetStatusAfterChanges | Status reflects all changes | R10 | ☑ PASS |
| testToString | toString() returns status | R10 | ☑ PASS |
| testMultiplePenChanges | Multiple pen state changes work | R3, R4 | ☑ PASS |
| testMultipleTurns | Multiple turns calculate correctly | R5, R6 | ☑ PASS |
| testStatusForAllDirections | Status shows all direction names | R10 | ☑ PASS |

**Total: 18 tests**

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### 6.3 Floor Tests (FloorTest.java)

| Test Case | Description | Requirement(s) | Result |
|---|---|---|---|
| testConstructorSize | Floor created with correct size | R1 | ☑ PASS |
| testConstructorInitializesZeros | All cells start as 0 | R1 | ☑ PASS |
| testConstructorInvalidSize | Size < 1 throws exception | R1, R16 | ☑ PASS |
| testConstructorValidSizes | Various valid sizes work | R1 | ☑ PASS |
| testMark | mark() sets cell to 1 | R8 | ☑ PASS |
| testMarkMultipleTimes | Marking same cell stays 1 | R8 | ☑ PASS |
| testMarkInvalidCoordinates | Invalid coords throw exception | R9, R16 | ☑ PASS |
| testGetValue | getValue() returns correct value | R8 | ☑ PASS |
| testGetValueInvalidCoordinates | Invalid coords throw exception | R9, R16 | ☑ PASS |
| testIsValidPositionTrue | Valid positions return true | R9 | ☑ PASS |
| testIsValidPositionFalse | Invalid positions return false | R9 | ☑ PASS |
| testReset | reset() clears all marks | R1 | ☑ PASS |

| Test Case | Description | Requirement(s) | Result |
|---|---|---|---|
| testPrint | print() returns string representation | R11 | ☑ PASS |
| testPrintMarkedCells | print() shows asterisks for marks | R11 | ☑ PASS |
| testPrintAxisLabels | print() includes axis labels | R11 | ☑ PASS |
| testGetGrid | getGrid() returns internal array | R8 | ☑ PASS |
| testFloorSizeOne | 1×1 floor works correctly | R1 | ☑ PASS |
| testCornerPositions | All corners accessible | R1, R8 | ☑ PASS |
| testMarkPattern | Multiple marks create pattern | R8 | ☑ PASS |

**Total: 23 tests (including parameterized variations)**

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### 6.4 Robot Simulator Tests (RobotSimulatorTest.java)

| Test Case | Description | Requirement(s) | Result |
|---|---|---|---|
| testInitialStateNotInitialized | Starts not initialized | R1 | ☑ PASS |
| testInitialStateRunning | Starts in running state | R14 | ☑ PASS |
| testInitializeCommand | "I 10" initializes 10×10 floor | R1 | ☑ PASS |
| testInitializeLowercase | "i 10" works (case insensitive) | R1, R15 | ☑ PASS |
| testInitializeResetsRobot | Initialize resets robot state | R2 | ☑ PASS |
| testInitializeInvalidSize | "I 0" returns error | R1, R16 | ☑ PASS |
| testInitializeNoArgument | "I" returns error | R16 | ☑ PASS |
| testInitializeInvalidArguments | Invalid args return error | R16 | ☑ PASS |
| testPenDown | "D" puts pen down | R4 | ☑ PASS |
| testPenUp | "U" puts pen up | R3 | ☑ PASS |
| testPenCommandsLowercase | "d"/"u" work | R3, R4, R15 | ☑ PASS |
| testTurnRight | "R" turns right | R5 | ☑ PASS |
| testTurnLeft | "L" turns left | R6 | ☑ PASS |
| testTurnCommandsLowercase | "r"/"l" work | R5, R6, R15 | ☑ PASS |
| testFourRightTurns | 4 rights = north | R5 | ☑ PASS |
| testMoveRequiresInit | Move before init returns error | R16 | ☑ PASS |
| testMoveNorth | Move north increases Y | R7 | ☑ PASS |
| testMoveEast | Move east increases X | R7 | ☑ PASS |

| Test Case | Description | Requirement(s) | Result |
|---|---|---|---|
| testMoveSouth | Move south decreases Y | R7 | ☑ PASS |
| testMoveWest | Move west decreases X | R7 | ☑ PASS |
| testMoveLowercase | "m 5" works | R7, R15 | ☑ PASS |
| testMoveNoSpace | "M5" works | R7 | ☑ PASS |
| testMoveWithPenDown | Marks floor when pen down | R8 | ☑ PASS |
| testMoveWithPenUp | No marks when pen up | R3 | ☑ PASS |
| testMoveStopsAtBoundary | Stops at floor edge | R9 | ☑ PASS |
| testMoveZeroSpaces | "M 0" does nothing | R7 | ☑ PASS |
| testMoveMissingArgument | "M" returns error | R16 | ☑ PASS |
| testStatusCommand | "C" returns status | R10 | ☑ PASS |
| testStatusReflectsChanges | Status updates correctly | R10 | ☑ PASS |
| testStatusLowercase | "c" works | R10, R15 | ☑ PASS |
| testPrintRequiresInit | Print before init returns error | R16 | ☑ PASS |
| testPrintCommand | "P" returns floor representation | R11 | ☑ PASS |
| testPrintShowsMarks | Print shows asterisks | R11 | ☑ PASS |
| testPrintLowercase | "p" works | R11, R15 | ☑ PASS |
| testQuitCommand | "Q" stops simulator | R14 | ☑ PASS |
| testQuitLowercase | "q" works | R14, R15 | ☑ PASS |
| testHistoryEmpty | Empty history message | R12 | ☑ PASS |
| testHistoryReplay | History replays commands | R13 | ☑ PASS |
| testHistoryExcludesCommands | P, H, Q not in history | R12 | ☑ PASS |
| testUnknownCommand | Unknown command returns error | R16 | ☑ PASS |
| testEmptyCommand | Empty command returns error | R16 | ☑ PASS |
| testNullCommand | Null command returns error | R16 | ☑ PASS |
| testWhitespaceCommand | Whitespace returns error | R16 | ☑ PASS |
| testCompleteScenario | Full integration test | R1-R16 | ☑ PASS |
| testDrawLShape | Draw L-shape pattern | R7, R8 | ☑ PASS |
| testCaseInsensitivity | All commands case insensitive | R15 | ☑ PASS |

**Total: 48 tests (including parameterized variations)**

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## 7. Test Execution Summary

| Test Class | Tests Run | Passed | Failed | Errors | Skipped | Time |
|---|---|---|---|---|---|---|
| DirectionTest | 30 | 30 | 0 | 0 | 0 | 0.245s |
| FloorTest | 23 | 23 | 0 | 0 | 0 | 0.073s |
| RobotSimulatorTest | 48 | 48 | 0 | 0 | 0 | 0.101s |
| RobotTest | 18 | 18 | 0 | 0 | 0 | 0.035s |
| **TOTAL** | **119** | **119** | **0** | **0** | **0** | **0.454s** |

Overall Result: ☑ BUILD SUCCESS

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## 8. Requirements Coverage Matrix

| Req ID | Requirement | Test Coverage | Status |
|---|---|---|---|
| R1 | Initialize N×N floor | testInitializeCommand, testConstructorSize, testConstructorValidSizes | ☑ Covered |
| R2 | Initial robot state (0,0), north, pen up | testInitialPosition, testInitialDirection, testInitialPenState, testInitializeResetsRobot | ☑ Covered |
| R3 | Pen up command | testPenUp, testMoveWithPenUp, testPenCommandsLowercase | ☑ Covered |
| R4 | Pen down command | testPenDown, testPenCommandsLowercase | ☑ Covered |
| R5 | Turn right command | testTurnRight, testTurnRightFrom*, testFourRightTurns | ☑ Covered |
| R6 | Turn left command | testTurnLeft, testTurnLeftFrom*, testFourLeftTurns | ☑ Covered |
| R7 | Move forward command | testMoveNorth/East/South/West, testMoveLowercase | ☑ Covered |
| R8 | Mark floor when pen down | testMoveWithPenDown, testMark, testMarkPattern | ☑ Covered |
| R9 | Boundary protection | testMoveStopsAtBoundary, testIsValidPosition* | ☑ Covered |
| R10 | Display robot status | testStatusCommand, testGetStatus*, testStatusReflectsChanges | ☑ Covered |
| R11 | Print floor with asterisks | testPrintCommand, testPrintShowsMarks, testPrintMarkedCells | ☑ Covered |

| Req ID | Requirement | Test Coverage | Status |
|---|---|---|---|
| R12 | Maintain command history | testHistoryExcludesCommands | ☑ Covered |
| R13 | Replay history | testHistoryReplay, testHistoryEmpty | ☑ Covered |
| R14 | Quit program | testQuitCommand, testQuitLowercase | ☑ Covered |
| R15 | Case insensitivity | testCaseInsensitivity, all *Lowercase tests | ☑ Covered |
| R16 | Error messages | testUnknownCommand, testEmptyCommand, testInitializeInvalidSize | ☑ Covered |

**Coverage: 16/16 requirements (100%)**

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## 9. Development Time Tracking

| Component/Function | Development Time | Test Development Time | Total | Notes |
|---|---|---|---|---|
| **Direction enum** | 30 min | 45 min | 1h 15min | Enum with behavior methods |
| **Robot class** | 45 min | 40 min | 1h 25min | State management |
| **Floor class** | 1h | 50 min | 1h 50min | Grid operations, print logic |
| **RobotSimulator** | 2h | 1h 30min | 3h 30min | Command parsing, integration |
| **Error handling** | 30 min | 30 min | 1h | Validation across all classes |
| **Documentation** | 1h | - | 1h | README, comments |
| **TOTAL** | **5h 45min** | **4h 15min** | **10h** | - |

### Time Distribution

- **Development:** 57.5%
- **Testing:** 42.5%

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## 10. How to Build and Run

### Prerequisites

- Java 11 or higher
- Maven 3.6 or higher

### Build Commands

```
# Compile the project
mvn compile

# Run all tests
mvn test

# Run tests with code coverage and threshold enforcement
mvn clean verify

# View JaCoCo HTML report (after running verify)
# Open: target/site/jacoco/index.html

# Package into executable JAR
mvn package

# Clean build artifacts
mvn clean
```

### Run the Application

```
# Using Maven
mvn exec:java

# Using JAR file
java -jar target/robot-simulator-1.0-SNAPSHOT.jar
```

---

## 11. Project Structure

```
robot-project/
├── pom.xml                          # Maven build configuration
├── README.md                        # Project documentation
├── docs/
│   └── REPORT.md                    # This report
└── src/
    ├── main/java/com/robot/
    │   ├── Direction.java           # Cardinal directions enum
    │   ├── Robot.java               # Robot state and behavior
    │   ├── Floor.java               # N×N grid representation
    │   └── RobotSimulator.java      # Main class with command loop
```

```
    └── test/java/com/robot/
        ├── DirectionTest.java          # 30 tests
        ├── RobotTest.java              # 18 tests
        ├── FloorTest.java              # 23 tests
        └── RobotSimulatorTest.java     # 48 tests
```

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## 12. AI Tools Declaration

### Tool Information

| Attribute | Details |
|---|---|
| **AI Tool Used** | Claude (Anthropic) via Cursor IDE |
| **Model Version** | Claude 3.5/Claude 4 |
| **Usage Period** | February 2026 |
| **Purpose** | Code assistance, design guidance, testing strategies |

### Development Prompts and AI Interactions

The following prompts were used during the development of this application. Each prompt represents a step in the iterative development process.

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**Prompt 1: Initial Design Discussion**

**User:**

> I need to build a Robot Simulator in Java. The robot walks around an N×N room, holds a pen that can be up or down. When pen is down, it traces shapes on the floor as it moves. What's a good object-oriented design approach for this?

**AI Response Summary:**

- Suggested separating concerns into distinct classes: `Direction` (enum), `Robot`, `Floor`, and `RobotSimulator`
- Recommended using an enum for directions to encapsulate turn logic
- Proposed keeping Robot unaware of Floor for better testability

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**Prompt 2: Direction Enum Implementation**

**User:**

> Can you help me implement the Direction enum? It should handle NORTH, EAST, SOUTH, WEST and have methods for turning left/right and getting movement deltas.

**AI Response Summary:**

- Provided Direction enum with `turnRight()`, `turnLeft()`, `getDeltaX()`, `getDeltaY()` methods
- Explained why behavior belongs in the enum rather than external switch statements

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**Prompt 3: Floor Class Design**

**User:**

> Now I need the Floor class. It should be an N×N grid where cells can be marked (1) or unmarked (0). How should I handle the coordinate system and printing?

**AI Response Summary:**

- Suggested using `int[][] grid` with validation
- Recommended `mark(x, y)` always sets to 1 (idempotent operation)
- Provided print logic with Y-axis inverted for visual representation

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**Prompt 4: Robot State Management**

**User:**

> For the Robot class, what state should it track and what methods does it need? Should it know about the floor boundaries?

**AI Response Summary:**

- Robot should track: x, y position, direction, penDown state
- Recommended Robot NOT know about Floor (Single Responsibility Principle)
- Boundary checking should happen in the coordinator (RobotSimulator)

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**Prompt 5: Command Processing**

**User:**

> How should I structure the command processing in RobotSimulator? I need to handle commands like I, U, D, R, L, M, P, C, H, Q.

**AI Response Summary:**

- Suggested switch-case pattern mapping commands to methods
- Recommended separating parsing from execution
- Provided error handling strategy for invalid commands

---

**Prompt 6: Movement Logic with Pen**

**User:**

> When the robot moves with pen down, how do I mark all cells along the path? Should I mark the starting position too?

**AI Response Summary:**

- Mark starting position if pen is down before moving
- Loop through each step, validate boundary, move robot, then mark if pen down
- Return error message if boundary would be exceeded

---

**Prompt 7: Unit Testing Strategy**

**User:**

> What unit tests should I write? How do I organize tests for good coverage?

**AI Response Summary:**

- Suggested test pyramid: unit tests for each class, integration tests for full scenarios
- Recommended testing: initial states, state transitions, boundary conditions, error cases
- Provided examples of parameterized tests for Direction enum

---

**Prompt 8: Testing Direction Enum**

**User:**

> Can you help me write comprehensive tests for the Direction enum?

**AI Response Summary:**

- Tests for each direction's display name
- Tests for turnRight/turnLeft from each direction
- Tests for delta values (movement offsets)
- Parameterized tests for properties that apply to all directions

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**Prompt 9: Testing Floor Class**

**User:**

> What edge cases should I test for the Floor class?

**AI Response Summary:**

- Constructor validation (size < 1 should throw)
- Boundary checking (isValidPosition for corners and edges)
- Mark idempotency (marking same cell twice)
- Reset functionality
- Print output verification

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**Prompt 10: Testing Robot Class**

**User:**

> Help me write tests for the Robot class covering all its behaviors.

**AI Response Summary:**

- Initial state tests (position, direction, pen state)
- State change tests (pen up/down, turns)
- Reset functionality
- Status string format verification

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**Prompt 11: Integration Testing**

**User:**

> I need integration tests that test the full command flow. Can you help with scenarios like drawing an L-shape?

**AI Response Summary:**

- Provided complete scenario test matching project description example
- L-shape drawing test with position and floor verification
- Case insensitivity test covering all commands

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**Prompt 12: Error Handling**

**User:**

> What error cases should I handle and test? I want meaningful error messages.

**AI Response Summary:**

- System not initialized errors
- Invalid command errors
- Missing/invalid arguments
- Boundary violation errors
- Null/empty input handling

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**Prompt 13: History Feature**

**User:**

> How should I implement the command history replay feature? Which commands should be excluded?

**AI Response Summary:**

- Store commands in ArrayList
- Exclude P (print), H (history), Q (quit) from history
- Replay by resetting state and re-executing all stored commands
- Display each command and its result during replay

---

**Prompt 14: Code Review and Refinement**

**User:**

> Can you review my implementation and suggest any improvements for code quality?

**AI Response Summary:**

- Suggested adding JavaDoc comments
- Recommended consistent error message formatting
- Verified Single Responsibility Principle adherence
- Confirmed test coverage is comprehensive

---

**Prompt 15: Documentation**

**User:**

> Help me create a README with build instructions and usage examples.

**AI Response Summary:**

- Provided README structure with project overview
- Build commands for Maven
- Command reference table
- Example session showing typical usage

---

## AI Contribution Summary

| Development Phase | AI Assistance Level | Description |
|---|---|---|
| Architecture Design | High | Object-oriented design patterns, class responsibilities |
| Core Implementation | Medium | Code structure, method implementations |
| Unit Testing | High | Test case identification, edge cases, parameterized tests |
| Error Handling | Medium | Error scenarios, message formatting |
| Documentation | Medium | README structure, report organization |
| Code Review | Low | Minor refinements and consistency checks |

## Declaration Statement

I declare that AI tools (specifically Claude via Cursor IDE) were used as a development assistant for this project. The AI provided guidance on:

- Object-oriented design principles
- Code structure and implementation patterns
- Unit test strategies and test case identification
- Documentation formatting

All code was reviewed, understood, and validated by the student before inclusion in the final submission. The AI served as a learning aid and productivity tool, not as a replacement for understanding the concepts.

---

# 13. Code Analysis with JaCoCo

## 13.1 Tool Selection

For code coverage analysis, we selected **JaCoCo (Java Code Coverage)** version 0.8.11. JaCoCo is the industry-standard code coverage tool for Java projects. It integrates seamlessly with Maven and JUnit 5, instruments bytecode at runtime using a Java agent, and produces detailed HTML, CSV, and XML reports. JaCoCo measures the following metrics that map directly to the assignment requirements:

| Assignment Metric | JaCoCo Counter | Description |
|---|---|---|
| (a) Function Coverage | METHOD | How many defined methods have been called |
| (b) Statement Coverage | INSTRUCTION | Number of bytecode instructions successfully executed |
| (c) Path Coverage | COMPLEXITY | Cyclomatic complexity paths exercised through control structures |
| (d) Condition Coverage | BRANCH | Boolean expressions (if/else, switch) validated |
| (e) Line Coverage | LINE | Source code lines executed |

## 13.2 Maven Configuration

The JaCoCo plugin was added to `pom.xml` with three execution goals:

1. **prepare-agent** – Instruments classes before test execution
2. **report** – Generates HTML/CSV/XML reports after tests run
3. **check** – Enforces minimum coverage thresholds (build fails if not met)

```xml
<plugin>
    <groupId>org.jacoco</groupId>
    <artifactId>jacoco-maven-plugin</artifactId>
    <version>0.8.11</version>
    <executions>
        <execution>
            <id>prepare-agent</id>
            <goals><goal>prepare-agent</goal></goals>
        </execution>
        <execution>
            <id>report</id>
            <phase>test</phase>
            <goals><goal>report</goal></goals>
        </execution>
        <execution>
```

```
                <id>check</id>
                <goals><goal>check</goal></goals>
                <configuration>
                    <rules>
                        <rule>
                            <element>BUNDLE</element>
                            <limits>
                                <limit>
                                    <counter>LINE</counter>
                                    <value>COVEREDRATIO</value>
                                    <minimum>0.80</minimum>
                                </limit>
                                <limit>
                                    <counter>BRANCH</counter>
                                    <value>COVEREDRATIO</value>
                                    <minimum>0.70</minimum>
                                </limit>
                                <limit>
                                    <counter>METHOD</counter>
                                    <value>COVEREDRATIO</value>
                                    <minimum>0.80</minimum>
                                </limit>
                                <limit>
                                    <counter>INSTRUCTION</counter>
                                    <value>COVEREDRATIO</value>
                                    <minimum>0.80</minimum>
                                </limit>
                                <limit>
                                    <counter>COMPLEXITY</counter>
                                    <value>COVEREDRATIO</value>
                                    <minimum>0.70</minimum>
                                </limit>
                            </limits>
                        </rule>
                    </rules>
                </configuration>
            </execution>
        </executions>
    </plugin>
```

To run coverage analysis:

```
mvn clean verify
```

Reports are generated at: `target/site/jacoco/index.html`

### 13.3 Predefined Code Coverage Threshold Values for Releasing

Before running the analysis, the following threshold values were established as **release gates**. The build will fail if any threshold is not met, preventing a release of insufficiently tested code.

| Metric | Counter | Threshold | Rationale |
|---|---|---|---|
| (a) Function Coverage | METHOD | **80%** | Most public methods should be exercised by tests |
| (b) Statement Coverage | INSTRUCTION | **80%** | Core logic instructions must be verified |
| (c) Path Coverage | COMPLEXITY | **70%** | Control flow paths should be well-exercised; lower threshold accounts for defensive/unreachable code |
| (d) Condition Coverage | BRANCH | **70%** | Boolean conditions should be tested both ways; lower threshold accounts for unreachable default branches |
| (e) Line Coverage | LINE | **80%** | The majority of source lines should be executed |

**Threshold Justification:**

- **80%** is the standard industry threshold for method, instruction, and line coverage. It ensures that the vast majority of code is tested while allowing for I/O methods (`main()`, `run()`) that require interactive input and are difficult to unit test.
- **70%** is used for branch and complexity coverage because defensive programming patterns (e.g., `default` cases in exhaustive switch statements on enums) create branches that are intentionally unreachable, making 80% difficult to achieve without artificial test contortions.

## 13.4 Code Coverage Results

**Overall Summary**

| Metric | Covered | Total | Coverage % | Threshold | Status |
|---|---|---|---|---|---|
| (a) Function/Method | 51 | 53 | **96.2%** | 80% | PASS |
| (b) Statement/Instruction | 820 | 883 | **92.9%** | 80% | PASS |
| (c) Path/Complexity | 100 | 114 | **87.7%** | 70% | PASS |
| (d) Condition/Branch | 90 | 105 | **85.7%** | 70% | PASS |
| (e) Line | 195 | 220 | **88.6%** | 80% | PASS |

**Build Result:** `[INFO] All coverage checks have been met.` — **BUILD SUCCESS**

**Per-Class Breakdown**

| Class | Instructions | Branches | Lines | Methods | Complexity |
|---|---|---|---|---|---|
| **Robot.java** | 100% (109/109) | 100% (2/2) | 100% (33/33) | 100% (15/15) | 100% (16/16) |
| **Floor.java** | 100% (187/187) | 100% (20/20) | 100% (33/33) | 100% (9/9) | 100% (19/19) |

| Class | Instructions | Branches | Lines | Methods | Complexity |
|---|---|---|---|---|---|
| **Direction.java** | 95.7% (90/94) | 87.5% (14/16) | 93.1% (27/29) | 100% (7/7) | 89.5% (17/19) |
| **RobotSimulator.java** | 88.0% (434/493) | 80.6% (54/67) | 81.6% (102/125) | 90.9% (20/22) | 80.0% (48/60) |

**JaCoCo HTML Report Screenshots**

**Overall Coverage Summary (target/site/jacoco/index.html):**

```
┌──────────────────────────────────────────────────────────────────────────┐
│  Robot Simulator - JaCoCo Coverage Report                                │
├──────────────┬──────────────────┬──────────┬──────────────────┬──────────┤
│ Element      │ Missed Instr.    │ Cov.     │ Missed Branches  │ Cov.     │
├──────────────┼──────────────────┼──────────┼──────────────────┼──────────┤
│ com.robot    │ 63 of 883        │ 92%      │ 15 of 105        │ 85%      │
├──────────────┼──────────────────┼──────────┼──────────────────┼──────────┤
│ Total        │ 63 of 883        │ 92%      │ 15 of 105        │ 85%      │
│              │ Missed: 14 Cxty  │ 114      │ Missed: 25 Lines │ 220      │
│              │ Missed: 2 Methods│ 53       │ Missed: 0 Classes│ 4        │
└──────────────┴──────────────────┴──────────┴──────────────────┴──────────┘
```

**Per-Class Coverage Detail (target/site/jacoco/com.robot/index.html):**

```
┌──────────────────────┬──────────────────┬──────────┬──────────────────┬──────────┐
│ Class                │ Missed Instr.    │ Cov.     │ Missed Branches  │ Cov.     │
├──────────────────────┼──────────────────┼──────────┼──────────────────┼──────────┤
│ RobotSimulator       │ 59 of 493        │ 88%      │ 13 of 67         │ 80%      │
│ Direction            │ 4 of 94          │ 95%      │ 2 of 16          │ 87%      │
│ Robot                │ 0 of 109         │ 100%     │ 0 of 2           │ 100%     │
│ Floor                │ 0 of 187         │ 100%     │ 0 of 20          │ 100%     │
└──────────────────────┴──────────────────┴──────────┴──────────────────┴──────────┘
```

## 13.5 Analysis of Uncovered Code

**Fully Covered Classes (100% across all metrics)**

**Robot.java** and **Floor.java** achieved 100% coverage on all metrics. Every method, branch, line, and instruction in these classes is exercised by the test suite.

**Direction.java — Minor Gaps (95.7% instruction, 87.5% branch)**

| Uncovered Code | Location | Reason |
|---|---|---|

| Uncovered Code | Location | Reason |
|---|---|---|
| `default: return this;` in `turnRight()` | Line 36 | Unreachable: all 4 enum values have explicit cases |
| `default: return this;` in `turnLeft()` | Line 50 | Unreachable: all 4 enum values have explicit cases |

**Assessment:** These `default` branches are **defensive dead code**. They exist as a safety net in case a new direction were added to the enum in the future. Since all four directions (NORTH, EAST, SOUTH, WEST) are explicitly handled, these branches can never be reached at runtime. **Decision: KEEP** — This is a best practice for enum switch statements.

**RobotSimulator.java — Larger Gaps (88.0% instruction, 80.6% branch)**

| Uncovered Code | Location | Reason |
|---|---|---|
| `run()` method | Lines 347-365 | Interactive I/O using `Scanner` — requires stdin input, not suitable for unit testing |
| `main()` method | Lines 372-374 | Application entry point — delegates to `run()` |
| `initialize()` size < 1 check | Line 77 | Redundant: already validated in `handleInitializeCommand()` at line 334 |
| `move()` spaces < 0 check | Lines 131-132 | Redundant: already validated in `handleMoveCommand()` at line 313 |
| `handleMoveCommand()` negative check | Lines 313-314 | Tests pass positive values; negative values caught by `handleInitializeCommand` logic flow |
| `handleMoveCommand()` catch block | Lines 317-318 | Test sends "I abc" which is caught by `handleInitializeCommand`; "M abc" is not explicitly tested |
| Partial branches in `replayHistory()` | Line 219 | The null/empty check — replay commands always return non-null, non-empty results |
| Partial branches in default case | Line 294 | The `addToHistory && !commandHistory.isEmpty()` — the `!addToHistory` path for unknown commands is not tested |

**Assessment by category:**

1. **`run()` and `main()` (2 methods, 19 lines):** These are **interactive I/O methods** that read from `System.in`. They cannot be practically unit tested without mocking stdin. All the logic they call (`processCommand()`) is fully tested. **Decision: KEEP — exclude from coverage expectations.**

2. **Redundant defensive validations (4 lines):** The `initialize()` method checks `size < 1` and `move()` checks `spaces < 0`, but these conditions are already caught by their respective handler methods (`handleInitializeCommand` and `handleMoveCommand`). These are defense-in-depth checks. **Decision: KEEP — defense-in-depth is good practice for public API methods.**

3. **Partial branch coverage in `replayHistory()` and default command (2 branches):** Edge cases where replay produces null/empty results or unknown commands are entered with `addToHistory=false`. **Decision: KEEP — these are valid defensive checks.**

---

# 14. Software Release Decision

## 14.1 Coverage vs. Threshold Summary

| Metric | Actual | Threshold | Margin | Verdict |
|---|---|---|---|---|
| (a) Function/Method | **96.2%** | 80% | +16.2% | **PASS** |
| (b) Statement/Instruction | **92.9%** | 80% | +12.9% | **PASS** |
| (c) Path/Complexity | **87.7%** | 70% | +17.7% | **PASS** |
| (d) Condition/Branch | **85.7%** | 70% | +15.7% | **PASS** |
| (e) Line | **88.6%** | 80% | +8.6% | **PASS** |

All five metrics exceed their predefined thresholds with comfortable margins ranging from +8.6% to +17.7%.

## 14.2 Qualitative Assessment

1. **Core Business Logic:** `Robot.java` and `Floor.java` — the core domain classes — have **100% coverage** on all metrics. This means every line, branch, method, and instruction in the robot movement and floor management logic has been verified.

2. **Command Processing:** `RobotSimulator.java` — the command processing engine — has **88%+ coverage** across all metrics. All 10 commands (I, U, D, R, L, M, P, C, H, Q) are tested with valid inputs, invalid inputs, boundary conditions, and case-insensitivity.

3. **Uncovered Code is Non-Critical:** The 25 uncovered lines are:

   - 19 lines in `run()`/`main()` — I/O scaffolding, not business logic
   - 4 lines of redundant defensive validations — duplicated safety checks
   - 2 lines of unreachable enum defaults — structural dead code

4. **No Uncovered Business Logic:** There are zero uncovered lines in any business logic path. Every user-facing feature is fully tested.

## 14.3 Release Decision

**RELEASE APPROVED**

| Criterion | Status |
|---|---|
| All 119 unit tests pass | **YES** |
| Function coverage ≥ 80% | **YES** (96.2%) |
| Statement coverage ≥ 80% | **YES** (92.9%) |

| Criterion | Status |
|---|---|
| Path coverage ≥ 70% | **YES** (87.7%) |
| Condition coverage ≥ 70% | **YES** (85.7%) |
| Line coverage ≥ 80% | **YES** (88.6%) |
| No uncovered business logic | **YES** |
| JaCoCo `check` goal passed | **YES** |

The software meets all predefined quality gates and is ready for release. The uncovered code consists entirely of interactive I/O methods, unreachable defensive code, and redundant safety validations — none of which represent risk to the application's correctness.

### 14.4 Recommendations for Future Improvement

1. **Consider removing redundant validations:** The `size < 1` check in `initialize()` and `spaces < 0` check in `move()` duplicate validation already done in their handler methods. Removing them would improve coverage metrics, though keeping them follows defense-in-depth principles.

2. **Integration testing for `run()` method:** Use `System.setIn()` and `ByteArrayInputStream` to provide scripted input for testing the interactive loop, which would cover the remaining 19 lines.

3. **Add test for `M abc` (invalid move argument):** This would cover the `NumberFormatException` catch block in `handleMoveCommand()` (lines 317-318).

---

## 15. AI Tool Interactions for Task 2 (Code Analysis)

### Prompt 16: Choosing a Code Coverage Tool

**User:**

> I need to do code coverage analysis for my Java project. What tools are available and which one should I use? I need to measure function coverage, statement coverage, path coverage, condition coverage, and line coverage.

**AI Response Summary:**

- Recommended **JaCoCo** as the industry-standard Java coverage tool
- Explained JaCoCo's counters map to the required metrics: METHOD → function, INSTRUCTION → statement, COMPLEXITY → path, BRANCH → condition, LINE → line
- Noted JaCoCo integrates with Maven via a plugin and generates HTML reports
- Alternative tools mentioned: Cobertura (outdated), Clover (commercial), PIT (mutation testing, different purpose)

---

### Prompt 17: Initial JaCoCo Setup

**User:**

> How do I add JaCoCo to my Maven pom.xml? I want it to run automatically during `mvn test`.

**AI Response Summary:**

- Provided the JaCoCo Maven plugin configuration with `prepare-agent` and `report` executions
- Explained that `prepare-agent` instruments the code and `report` generates HTML/CSV/XML
- Showed how to run with `mvn clean test` and find reports at `target/site/jacoco/`

---

### Prompt 18: First Coverage Run — Analyzing Results

**User:**

> I ran the coverage and got 92% instruction coverage and 85% branch coverage. Some lines in RobotSimulator are not covered — the `run()` method and `main()`. Is this a problem?

**AI Response Summary:**

- Explained that `run()` and `main()` are I/O methods that read from `System.in` — they cannot be practically unit tested without mocking stdin
- The important thing is that all business logic called by these methods (via `processCommand()`) is fully tested
- Suggested this is normal and acceptable — focus coverage efforts on business logic, not I/O scaffolding
- Noted that 92% overall coverage is excellent for a project of this nature

---

### Prompt 19: Understanding Uncovered Branches

**User:**

> JaCoCo shows some branches missed in Direction.java — the `default` cases in the switch statements. But I test all four directions. Why are these uncovered?

**AI Response Summary:**

- Explained that `default` branches in exhaustive enum switch statements are unreachable dead code
- Java requires a `default` case for completeness, but since all enum values are explicitly handled, the default can never execute
- This is a known JaCoCo limitation — it reports branches that are structurally present but semantically unreachable
- Recommended keeping the `default` cases as defensive coding practice but excluding them from coverage concerns

---

### Prompt 20: Setting Coverage Thresholds

**User:**

> What are good threshold values for code coverage to use as release gates? Should I use the same threshold for all metrics?

**AI Response Summary:**

- Recommended **80%** for line, method, and instruction coverage as the industry standard
- Recommended **70%** for branch and complexity coverage due to unreachable defensive code inflating the denominator
- Explained that thresholds should be achievable without test contortions but still meaningful
- Showed how to add the `check` execution to JaCoCo with threshold rules
- Noted that the Maven build will fail if thresholds are not met, acting as a quality gate

---

### Prompt 21: Configuring Release Enforcement

**User:**

> Can you help me configure JaCoCo to fail the build if coverage drops below the thresholds? I want this to be enforced automatically.

**AI Response Summary:**

- Added the `check` execution to the JaCoCo plugin configuration
- Configured five limits: LINE ≥ 80%, BRANCH ≥ 70%, METHOD ≥ 80%, INSTRUCTION ≥ 80%, COMPLEXITY ≥ 70%
- Explained the `BUNDLE` element applies thresholds at the project level (not per-class)
- Showed that running `mvn clean verify` triggers all three phases: agent → report → check
- Verified build output: `[INFO] All coverage checks have been met.`

---

### Prompt 22: Analyzing the Final Results for the Report

**User:**

> The build passed with all checks met. Can you help me write up the analysis for the report? I need to discuss which code is covered, which isn't, and whether the uncovered code should stay or be removed.

**AI Response Summary:**

- Categorized uncovered code into three groups: I/O methods, defensive dead code, and redundant validations
- Recommended keeping all uncovered code with clear justifications for each category
- Noted that 100% coverage on Robot.java and Floor.java (core domain) is the most important finding
- Suggested that the release is approved since all thresholds are met and no business logic is uncovered
- Provided recommendations for future improvement (integration tests for `run()`, test for `M abc`)

---

## 16. Conclusion

The Robot Simulator project successfully implements all 16 functional requirements and meets all code coverage quality gates:

### Task 1 Achievements

- ☑ **119 unit tests** — all passing
- ☑ **100% requirements coverage** — all 16 requirements verified
- ☑ **Clean, modular architecture** following SOLID principles
- ☑ **Comprehensive error handling** with meaningful messages
- ☑ **Case-insensitive command processing**

### Task 2 Achievements

- ☑ **JaCoCo code coverage** integrated into Maven build
- ☑ **96.2% function coverage** (threshold: 80%)
- ☑ **92.9% statement coverage** (threshold: 80%)
- ☑ **87.7% path coverage** (threshold: 70%)
- ☑ **85.7% condition coverage** (threshold: 70%)
- ☑ **88.6% line coverage** (threshold: 80%)
- ☑ **Automated quality gates** — build fails if coverage drops below thresholds
- ☑ **Release approved** — all metrics exceed predefined thresholds

The application can be downloaded, compiled, tested, and verified using standard Maven commands:

```
mvn clean verify    # Compile, test, generate coverage, and enforce thresholds
```

---

*Report generated for COEN-6761 course project submission.*